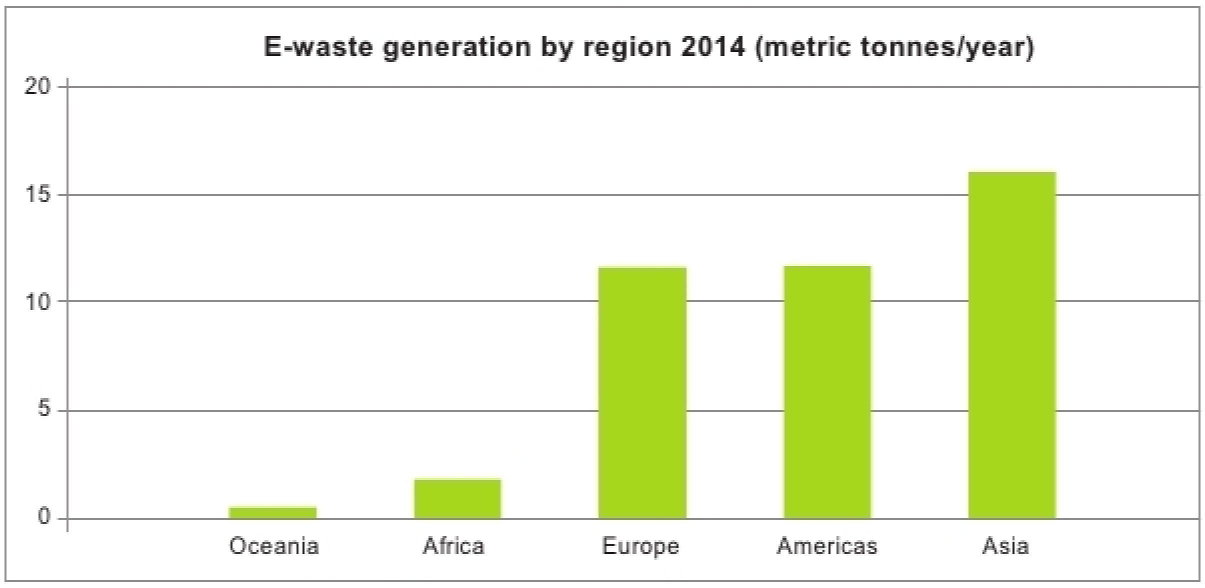

Figure 1: Increasing e-waste trend world over (Credit: UNU - The Global E-Waste Monitor 2014)

would also indirectly provide a solution for reducing particles in vibration.

In the process of minimisation, two more revolutionary technologies have emerged: molecular scale of electronics (MSE) and quantum computing. It was the quest for an ever decreasing size, but more complex electronic components with high speed ability, which gave rise to MSE.

The concept that molecules may be designed to operate as a self-contained device was put forwarded by Forrest L. Carter. He proposed some molecular components analogous of conventional electronic switches, gates and connections. Accordingly, the idea of a molecular P-N junction emerged. MSE is a simple interpolation of IC scaling.

Scaling is an attractive technology. Scaling of FET and MOS transistors is more rigorous and well defined than that of bipolar transistors. But there are problems in scaling of silicon technology. In scaling, while propagation delay should be minimum and packing density should be high, these should not be at the expense of the power dissipated. With these scaling rules in minds, scaling technology of silicon is reaching a limit.

Dr Barker reported that, "change, spin, conformation, colour, reactivity and lock-and-key recognition are just a few examples of molecular properties, which might be useful for representing and transforming logical information. To be useful, molecular scale logic will have to function close to the information theoretical limit of one bit on one carrier. Experimental practicalities suggest that it will be too easy to construct regular molecular arrays, preferably by chemical and physical self-organisation. This suggests that the natural logic architectures should be cellular automata: regular arrays of locally connected finite state machines where the state of each molecule might be represented by colour or by conformation. Schemes such as spectral hole burning already exist for storing and retrieving information in molecular arrays using light. The general problem of interfacing to a molecular system remains problematic. Molecular structures may be the first to take practical advantages of novel logic concepts such as emergent computation and 'floating architecture' in which computation is viewed as a self-organising process in a fluid-like medium."

Change is the only thing that is permanent in the universe. In technology scenario, changes become inevitable means of evolution and revolution. In tune, a new generation of IT known as Quantum Computing (QC) has come up. Mechanical computing, electronic computing, quantum computing, DNA computing, cloud computing, chemical computing and bio computing are a few generation-wise migrations of information technology (IT).

In the conventional computers we work with now, computing and processing of data is based on transistors' on and off states as binary representation of '1' or '0' or vice versa. In quantum computers, the basic principle is to use quantum properties to represent data. Here, computation and processing of data is made with quantum mechanical phenomena such as superposition, parallelism and entanglement. Therefore, whereas in conventional computers data is represented by binary 'bits,' in quantum computers representation is done with 'qubits' (quantum bits).

Qubits are typically subatomic particles such as electrons and photons. Generating, processing and managing qubits is an engineering challenge. Superiority of quantum computing over classical computing is multi-fold. First, whereas in classical computing logical bits are represented by on and off states of transistors, in quantum computing qubits are harnessed by properties of subatomic particles. Size of quantum computers will thus be much smaller than that of present-day computers. Both MSE and QC are thus found to be indirect solutions for green computing.

At current state of technology march, green ICT may be better looked at as a challenge to realise eco-friendly and environmentally-responsible solutions in order to reduce just not heat dissipation but also to maximise energy efficiency, recyclability and bio-degradability. Fact is, fast-growing production of electrical, electronic and computing equipment has resulted in enormous increase of e-waste, and especially carbon dioxide, which is responsible for creating havoc in the environment and for increasing pollution. As per a report published by International Telecommunication Union (ITU), e-waste has increased rapidly and reached a global high. Increasing trend of e-waste all over the world is shown in Figure 1.

Many studies have established that computers and IT industries dissipate